

(1) You have to complete the cheatsheet in your own handwriting in coloured ink - no black ink may be used. (2) No problems or solutions to problems or examples from a textbook may be written on this cheatsheet. (3) No derivation of equations may be written on this cheatsheet. (4) Do not fold this cheatsheet; a folded cheatsheet will be confiscated. (5) You have to hand in your cheatsheet with your test - we will subtract 10% if there is no cheatsheet with your test. (6) Complete your cheatsheet in the space left below AND on the back side of the page (A4 paper size). (7) You may only have ONE cheatsheet in the exam venue.

Kinematics (const a) $v_t = v_i + at$ $x_f = x_i + v_i t + \frac{1}{2} at^2$ $v_f^2 = v_i^2 + 2a(\Delta x)$ $x_f = x_i + \frac{1}{2} (v_i + v_f)t$ Projectile (2d) $v_{x_i} = v_i \cos \theta \quad x_i^* = 0$ $v_{y_i} = v_i \sin \theta \quad y_i^* = 0$ $\Delta x = v_{x_i} t + \frac{1}{2} a_x t^2$ $\Delta y = v_{y_i} t + \frac{1}{2} a_y t^2$ $t_u = \frac{v_{y_i}}{g} = \frac{v_i \sin \theta}{g}$ $y_{it} = y_{max} = v_{y_i} t_u - \frac{1}{2} g t_u^2$ $= \frac{v_i^2 \sin^2 \theta}{2g}$ $t_R = 2t_u$ $x_R = v_{x_i} t_R = \frac{v_i^2 \sin 2\theta}{g}$ Relativity $\vec{r}_{PA} = \vec{r}_{PB} + \vec{r}_{BA}$ $\vec{u}_{PA} = \vec{u}_{PB} + \vec{v}_{BA}$ $\vec{a}_{PA} = \vec{a}_{PB}$ Newton's $1. \vec{F}_{net} = \sum \vec{F}_i = 0$ $2. \vec{F}_{net} = \sum \vec{F}_i = m\vec{a}$ $3. \vec{F}_{2,1} = \vec{F}_{1,2}$ $f_s, k = N \mu_s, k$ $W = mg = F_g$ Circular motion Uniform: $a_c = \frac{v^2}{r} = r\omega^2$ $T = \frac{2\pi r}{v}$ $\omega = \frac{2\pi}{T} = \frac{v}{r}$ $\sum \vec{F}_r = m a_c = m \frac{v^2}{r}$ Non-Uniform: $\vec{a} = \vec{a}_r + \vec{a}_t$ $\sum \vec{F} = \sum \vec{F}_r + \sum \vec{F}_t$ $T = mg \left(\frac{v^2}{Rg} + \cos \theta \right)$ +1; top -1; bot	Energy $W = F \cos \theta$ $W_{net} = \int_{x_i}^{x_f} F_{net} dx$ $W = \vec{F} \cdot \Delta \vec{r}$ $W_{ext} = \sum W = \int \sum F_x d\vec{r}$ $\vec{F}_s = -kx = -F_{ext}$ $W_s = \frac{1}{2} kx_i^2 - \frac{1}{2} kx_f^2$ $W_s = \int_{x_i}^{x_f} (-kx) dx = -W_{ext}$ $\Delta E_{sys} = W_{ext}$ $\Delta K + \Delta U_g + \Delta U_s + \Delta E_{int} = W_{ext}$ Kinetic Energy: $W_{net} = \Delta K$ $K = \frac{1}{2} m v^2$ $K_f = K_i + W_{net}$ Potential Energy: $W_{ext} = \Delta U$ $U_{grav} = mgy$ $U_s = \frac{1}{2} kx^2$ Conservative forces $W_{con} = -\Delta U = W_{int}$ $E_{mech} = 0$ $F_x = \frac{dU(x)}{dx} = -kx$ Non conservative $E_{mech} = K + U$ Grav PE: $F_{grav}(r) = -\frac{GMm}{r^2}$ $U_{grav}(r) = -\frac{Gm_1 m_2}{r}$ Conservation Non-isolated $\Delta E_{sys} = W_{ext}$ Isolated $\Delta K + \Delta U = 0$ $\Delta E_{mech} = 0$ $\Delta E_{sys} = 0$ $\sum \vec{F}_{ext} = M \vec{a}_{cm}$	Power $P_{avg} = \frac{W}{\Delta t}$ $P = \frac{dW}{dt} = \vec{F} \cdot \vec{v}$ Kinetic friction $\sum W_{other} = f_k d = \Delta K$ $\Delta E_{int} = f_k d$ $\Delta E_{mech} = \Delta K + \Delta U_g = -f_k d$ $\Delta K + \Delta U + \Delta E_{int} = 0$ Linear momentum Isolated: $\frac{d(m_1 \vec{v}_1 + m_2 \vec{v}_2)}{dt} = 0$ $\vec{p} = m \vec{v}$ $\vec{p}_{total} = \vec{p}_1 + \vec{p}_2$ $\Delta \vec{p}_{total} = 0$ $\sum \vec{F} = \frac{d\vec{p}}{dt}$ Non-isolated $\Delta \vec{p} = \int_{t_i}^{t_f} \sum \vec{F} dt = \vec{I}$ $\vec{I} = \sum \vec{F} \Delta t \text{ (const } \sum \vec{F})$ Inelastic collision $\Delta \vec{p} = 0$ $\vec{v}_f = \frac{m_1 \vec{v}_{1,i} + m_2 \vec{v}_{2,i}}{m_1 + m_2}$ $\vec{v}_f = \frac{m_1}{m_1 + m_2} \vec{v}_{1,i} \quad (\vec{v}_{2,i} = 0)$ Elastic collision $\Delta \vec{p} = 0 \quad \Delta K = 0$ $\vec{v}_{1,f} = \left(\frac{m_1 - m_2}{m_1 + m_2} \right) \vec{v}_{1,i} + \left(\frac{2m_2}{m_1 + m_2} \right) \vec{v}_{2,i}$ $\vec{v}_{2,f} = \left(\frac{2m_1}{m_1 + m_2} \right) \vec{v}_{1,i} + \left(\frac{m_1 - m_2}{m_1 + m_2} \right) \vec{v}_{2,i}$ $\vec{v}_{1,f} = \left(\frac{m_1 - m_2}{m_1 + m_2} \right) \vec{v}_{1,i} \quad (\vec{v}_{2,i} = 0)$ $\vec{v}_{2,f} = \left(\frac{2m_1}{m_1 + m_2} \right) \vec{v}_{1,i} \quad (\vec{v}_{2,i} = 0)$ CoM $\vec{r}_{cm} = \frac{1}{M} \sum m_i \vec{r}_i$ Many particles $\vec{v}_{cm} = \frac{d\vec{r}_{cm}}{dt} = \frac{1}{M} \sum m_i \vec{v}_i$ $\vec{p}_{total} = M \vec{v}_{cm} = \sum m_i \vec{v}_i$ $\vec{a}_{cm} = \frac{d\vec{v}_{cm}}{dt} = \frac{1}{M} \sum m_i \vec{a}_i$	RRO $s = r\theta$ $v = r\omega$ $a_t = r\alpha$ $\omega_{avg} = \frac{\Delta \theta}{\Delta t}; \omega = \frac{d\theta}{dt}$ $\alpha_{avg} = \frac{\Delta \omega}{\Delta t}; \alpha = \frac{d\omega}{dt}$ $\omega_f = \omega_i + \alpha t$ $\theta_f = \theta_i + \omega_i t + \frac{1}{2} \alpha t^2$ $\omega_f^2 = \omega_i^2 + 2\alpha(\theta_f - \theta_i)$ $\theta_f = \theta_i + \frac{1}{2} (\omega_i + \omega_f) t$ Torque ($\vec{\tau}$) $\vec{\tau} = \vec{r} \times \vec{F} \sin \theta = Fd$ $\vec{\tau} = \vec{r} \times \vec{F} = rF \sin \theta$ $\vec{\tau} > 0 \text{ (ccw)}; \vec{\tau} < 0 \text{ (cw)}$ RRO under a net force $\sum \vec{F}_t = m \vec{a}_t$ $\vec{\tau} = \tau_{net} = (\sum \vec{F}_t) r = I \alpha$ $I = \int r^2 dm = mr^2$ Rotational KE $K_r = \sum K_i = \frac{1}{2} I \omega^2$ RRO Energy cons. $ ds = r d\theta$ $dW = \vec{F} \cdot d\vec{s}$ $dW = \tau d\theta$ $P = \tau \omega$ $W_{total} = \Delta K_r$ $\Delta E_{sys} = W_{ext}$ $\Delta K + \Delta U + \Delta K_r + \Delta K_t + \Delta E_{int} = W_{ext}$ Moments of Inertia (I_{cm}) Thin cylinder: $I = MR^2$ Hollow Hollow cylinder: $I = \frac{1}{2} M(R_1^2 + R_2^2)$ Solid cylinder: $I = \frac{1}{2} MR^2$ Rectangular plate: $\frac{1}{12} M(a^2 + b^2)$ Rod centre: $\frac{1}{12} ML^2$ Rod end: $\frac{1}{3} ML^2$ Solid sphere: $\frac{2}{5} MR^2$ Hollow sphere: $\frac{2}{3} MR^2$
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SHM spring

$$F_s = -kx$$

$$a_x = -\frac{k}{m}x$$

$$\frac{d^2x}{dt^2} = -\omega^2 x$$

$$\omega = \sqrt{\frac{k}{m}}$$

$$x(t) = A \cos(\omega t + \phi)$$

$$v(t) = -A\omega \sin(\omega t + \phi) = \frac{dx}{dt}$$

$$a(t) = -A\omega^2 \cos(\omega t + \phi) = \frac{d^2x}{dt^2}$$

$$v_{\max} = A\omega = \sqrt{\frac{k}{m}} A$$

$$a_{\max} = A\omega^2 = \frac{k}{m} A$$

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{m}{k}}$$

$$f = \frac{1}{T} = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

Energy SHO

$$\Delta E_{\text{mech}} = 0$$

$$E_{\text{mech}} = K + U$$

$$K = \frac{1}{2} k A^2 \sin^2(\omega t + \phi)$$

$$U_{\text{spring}} = \frac{1}{2} k A^2 \cos^2(\omega t + \phi)$$

$$E_{\text{mech}} = \frac{1}{2} k A^2$$

Pendulum (simple)

$$\Sigma F_t = m a_t$$

$$a_t = \frac{d^2s}{dt^2}$$

$$s = L\theta$$

$$\frac{d^2\theta}{dt^2} = -\frac{g}{L} \theta \quad \theta < 10^\circ$$

$$\frac{d^2\theta}{dt^2} = -\omega^2 \theta$$

$$\omega^2 = \frac{g}{L} \quad T = 2\pi \sqrt{\frac{L}{g}}$$

$$\theta = \theta_{\max} \cos(\omega t + \phi)$$

Pendulum (physical)

$$\tau = -mgd \sin \theta = I \alpha$$

(Newton's) 2nd law

$$\frac{d^2\theta}{dt^2} = -\frac{mgd}{I} \sin \theta \approx -\frac{mgd}{I} \theta$$

$$= -\frac{mgd}{I} \theta = -\omega^2 \theta$$

$$\omega = \sqrt{\frac{mgd}{I}}$$

$$T = 2\pi \sqrt{\frac{I}{mgd}}$$

Wave motion

$$y(x, t) = f(x \pm vt)$$

$$y(x, t) = A \sin \left[\frac{2\pi}{\lambda} (x - vt) \right]$$

$$k = \frac{2\pi}{\lambda}$$

$$\omega = \frac{2\pi}{T} = 2\pi f$$

$$y = A \sin(kx - \omega t + \phi)$$

$$v = \lambda f = \frac{\omega}{k}$$

$$y_{\max} = \omega A$$

$$a_{y, \max} = \omega^2 A$$

Superposition

$$y(x, t) = 2A \cos\left(\frac{\phi}{2}\right) \sin(kx - \omega t + \frac{\phi}{2})$$

constructive $\rightarrow \phi = \text{even multiple of } \pi$

destructive $\rightarrow \phi = \text{odd multiples of } \pi$

Units

Frequency f Hz s^{-1}

Joule J W/E/A J $kg m^2/s^2$ Nm

Force $\vec{F}$ N $kg m/s^2$

Power P W $kg m^2/s^3$

Pressure p Pa $kg/m s^2$ N/m²

Density ρ kg/m^3

Momentum $\vec{p}$ $kg m/s$

Impulse $\vec{I}$ N s

Torque $\vec{\tau}$ Nm $kg m^2/s^2$

Energy E $kg m^2/s^2$

Wave number k m^{-1}

spring constant k N/m

Standing waves

$$y(x, t) = 2A \sin(kx) \cos(\omega t)$$

Nodes: $\sin(kx) = 0$

$$x = \frac{n\lambda}{2}, n = 0, 1, 2, \dots$$

Anti-nodes: $\sin(kx) = \pm 1$

$$x = \frac{n\lambda}{4}, n = 1, 3, 5, \dots$$

Dimensions

metres L m

mass M kg

time T s

Temperature

$$T_C = T - 273.15$$

$$\alpha = \frac{\Delta L}{L_i}$$

$$\alpha = \frac{\Delta L / L_i}{\Delta T}$$

$$\Delta L = \alpha L_i \Delta T$$

$$\Delta V = \beta V_i \Delta T$$

$$\beta = 3\alpha$$

1st law thermodynamics

$$Q = mc \Delta T$$

$$T_C < T_F < T_H$$

$$-Q_{\text{hot}} = +Q_{\text{cold}} \rightarrow -m_h c_h (T_f - T_h) = m_c c_c (T_f - T_c)$$

$$\Delta E_{\text{int}} = 0 \text{ (isolated system)}$$

Latent heat

$$Q = L \Delta m$$

$$\Delta E_{\text{int}} = Q + W$$

$$W = nRT \ln \left(\frac{V_i}{V_f} \right)$$

$$W = - \int_{V_i}^{V_f} P dV = -P(V_f - V_i)$$

$$P = \frac{|\vec{F}_{\text{net}}|}{A}$$